

Application Packaging, Scripting, CMD, PowerShell, Bash & InstallShield Learning Roadmap

A beginner-to-professional learning guide for Application Packaging Engineers.

Phase 1: Windows Fundamentals

- Windows File System: drives, files, folders, Program Files, ProgramData, AppData, Windows directory structure.
- Windows Registry: hives (HKLM, HKCU, HKCR, HKU, HKCC), keys, values, permissions, export/import operations.
- Windows Services: service lifecycle, startup types, management using Services.msc and SC commands.

Phase 2: Command Prompt (CMD)

- Navigation commands: cd, dir, tree, cls.
- File operations: copy, xcopy, robocopy, move, del, rename, type.
- Folder operations: mkdir, rmdir.
- Networking: ping, ipconfig, tracert, nslookup.
- Process management: tasklist, taskkill.
- Environment variables: set, setx, echo.
- Registry commands: reg query, reg add, reg delete, reg export.

Phase 3: Batch Scripting

- Script structure and execution.
- Variables and user input.
- IF statements and conditional logic.
- FOR loops and iteration.
- Functions using labels and call.
- Error handling using ERRORLEVEL.
- Automation exercises for installers and deployments.

Phase 4: PowerShell

- PowerShell basics and help system.
- Variables, arrays, hashtables, objects.
- Conditions and loops.
- Functions and reusable scripts.
- File and folder automation.
- Registry management.
- Service and process management.

- WMI/CIM queries.
- Error handling, logging, modules and scripting best practices.

Phase 5: Bash Scripting

- Linux filesystem navigation.
- Variables, conditions, loops and functions.
- grep, find, awk basics.
- Shell automation and script execution.
- Permissions and file ownership.

Phase 6: MSI Fundamentals

- MSI architecture.
- Features, Components, Product Code, Package Code, Upgrade Code.
- MSI database tables.
- MSIEXEC installation, uninstall and logging commands.
- Understanding silent installation techniques.

Phase 7: InstallShield

- InstallShield project types.
- Files and Folders view.
- Features and Components.
- Registry, shortcuts and environment variables.
- Custom Actions.
- Build configuration and release creation.
- Troubleshooting package creation.

Phase 8: Packaging Tools

- Orca MSI editor.
- Process Monitor analysis.
- RegShot comparison.
- Installation capture and troubleshooting techniques.

Phase 9: Deployment Tools

- SCCM/MECM applications and packages.
- Detection methods and deployment strategies.
- Intune Win32 applications.
- Detection rules, requirements and PowerShell deployment scripts.

Phase 10: Advanced Topics

- MST transforms.

- Merge modules.
- COM and DLL registration.
- Active Setup.
- Upgrade scenarios.
- Patch creation.
- Advanced PowerShell for enterprise packaging.

Recommended Learning Schedule

Month 1: Windows Fundamentals + CMD

Month 2: Batch Scripting + PowerShell Basics

Month 3: Registry + MSI Fundamentals + Orca

Month 4: InstallShield Projects and Custom Actions

Month 5: ProcMon + SCCM + Intune

Month 6: Advanced Packaging and Automation

Interview Focus Areas

- MSI vs EXE
- Product Code vs Upgrade Code
- Components and Features
- Silent Installations
- Custom Actions
- MSI Logging
- Registry Troubleshooting
- ProcMon Analysis
- PowerShell Scripting
- Intune and SCCM Deployments